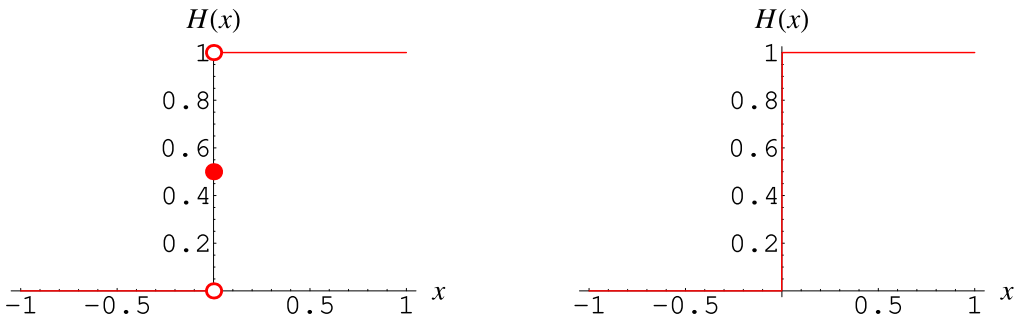
Heaviside Step Function

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The Heaviside step function is a mathematical function denoted $H(x)$, or sometimes $\theta(x)$ or $u(x)$ (Abramowitz and Stegun 1972, p. 1020), and also known as the "unit step function." The term "Heaviside step function" and its symbol can represent either a piecewise constant function or a generalized function.



When defined as a piecewise constant function, the Heaviside step function is given by

$$H(x)=\begin{cases}0 & x<0\\ \frac{1}{2} & x=0\\ 1 & x>0\end{cases} \tag{1}$$

(Abramowitz and Stegun 1972, p. 1020; Bracewell 2000, p. 61). The plot above shows this function (left figure), and how it would appear if displayed on an oscilloscope (right figure).

When defined as a generalized function, it can be defined as a function $\theta(x)$ such that

$$\int \theta(x)\,\phi'(x)\,dx=-\phi(0) \tag{2}$$

for $\phi'(x)$ the derivative of a sufficiently smooth function $\phi(x)$ that decays sufficiently quickly (Kanwal 1998).

The Wolfram Language represents the Heaviside generalized function as HeavisideTheta, while using UnitStep to represent the piecewise function Piecewise[{{1, x >= 0}}] (which, it should be noted, adopts the convention $H(0)=1$ instead of the conventional definition $H(0)=1/2$).

The shorthand notation

$$H_c(x) \equiv H(x-c) \tag{3}$$

is sometimes also used.

The Heaviside step function is related to the boxcar function by

$$\Pi(x) = H\left(x+\tfrac{1}{2}\right) - H\left(x-\tfrac{1}{2}\right) \tag{4}$$

and can be defined in terms of the sign function by

$$H(x) = \tfrac{1}{2}\left[1 + \operatorname{sgn}(x)\right]. \tag{5}$$

The derivative of the step function is given by

$$\frac{d}{dx} H(x) = \delta(x), \tag{6}$$

where $\delta(x)$ is the delta function (Bracewell 2000, p. 97).

The Heaviside step function is related to the ramp function $R(x)$ by

$$R(x) = x H(x), \tag{7}$$

and to the derivative of $R(x)$ by

$$\frac{d}{dx} R(x) = H(x). \tag{8}$$

The two are also connected through

$$R(x) = H(x) * H(x), \tag{9}$$

where $*$ denotes convolution.

Bracewell (2000) gives many identities, some of which include the following. Letting $*$ denote the convolution,

$$H(x) * f(x) = \int_{-\infty}^{x} f(x')\,dx' \tag{10}$$

$$H(t) * H(t) = \int_{-\infty}^{\infty} H(u)\,H(t-u)\,du \tag{11}$$

$$= H(0) \int_{0}^{\infty} H(t-u)\,du \tag{12}$$

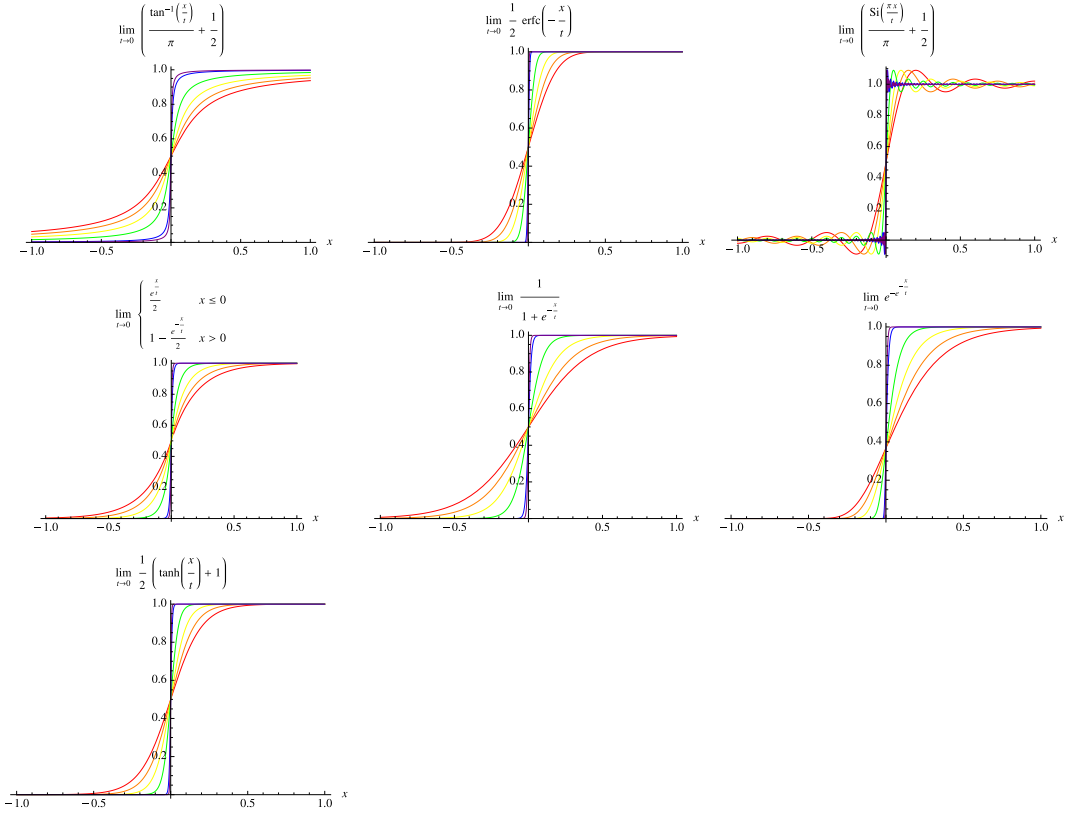
$$= H(0)\,H(t) \int_{0}^{t} du \tag{13}$$

$$= t\,H(t). \tag{14}$$

In addition,

$$H(ax+b) = H\left(x+\frac{b}{a}\right) H(a) + H\left(-x-\frac{b}{a}\right) H(-a) \tag{15}$$

$$= \begin{cases} H\left(x+\dfrac{b}{a}\right) & a > 0 \\ H\left(-x-\dfrac{b}{a}\right) & a < 0. \end{cases} \tag{16}$$



The Heaviside step function can be defined by the following limits,

$$H(x) = \lim_{t\to 0}\left[\frac{1}{2} + \frac{1}{\pi}\tan^{-1}\left(\frac{x}{t}\right)\right] \tag{17}$$

$$= \frac{1}{\sqrt{\pi}}\lim_{t\to 0}\int_{-x}^{\infty} t^{-1}\, e^{-u^2/t^2}\, du \tag{18}$$

$$= \frac{1}{2}\lim_{t\to 0}\operatorname{erfc}\left(-\frac{x}{t}\right) \tag{19}$$

$$= \frac{1}{\pi}\lim_{t\to 0}\int_{-\infty}^{x} t^{-1}\operatorname{sinc}\left(\frac{u}{t}\right) du \tag{20}$$

$$= \frac{1}{\pi}\lim_{t\to 0}\int_{-\infty}^{x}\frac{1}{u}\sin\left(\frac{u}{t}\right) du \tag{21}$$

$$= \frac{1}{2} + \frac{1}{\pi}\lim_{t\to 0}\operatorname{si}\left(\frac{\pi x}{t}\right) \tag{22}$$

$$= \lim_{t\to 0}\begin{cases}\frac{1}{2}\, e^{x/t} & \text{for } x \le 0\\ 1 - \frac{1}{2}\, e^{-x/t} & \text{for } x \ge 0\end{cases} \tag{23}$$

$$= \lim_{t\to 0}\frac{1}{1+e^{-x/t}} \tag{24}$$

$$= \lim_{t\to 0} e^{-e^{-x/t}} \tag{25}$$

$$= \frac{1}{2}\lim_{t\to 0}\left[1 + \tanh\left(\frac{x}{t}\right)\right] \tag{26}$$

$$= \lim_{t\to 0}\int_{-\infty}^{x} t^{-1}\,\Lambda\left(\frac{x-\frac{1}{2}\,t}{t}\right) dx, \tag{27}$$

where $\operatorname{erfc}(x)$ is the erfc function, $\operatorname{si}(x)$ is the sine integral, $\operatorname{sinc}(x)$ is the sinc function, and $\Lambda(x)$ is the one-argument triangle function. The first four of these are illustrated above for $t = 0.2$, 0.1, and 0.01.

Of course, any monotonic function with constant unequal horizontal asymptotes is a Heaviside step function under appropriate scaling and possible reflection. The Fourier transform of the Heaviside step function is given by

$$\mathcal{F}[H(x)] = \int_{-\infty}^{\infty} e^{-2\pi i k x} H(x)\, dx \tag{28}$$

$$= \frac{1}{2}\left[\delta(k) - \frac{i}{\pi k}\right], \tag{29}$$

where $\delta(k)$ is the delta function.

SEE ALSO

Absolute Value, Boxcar Function, Delta Function, Fourier Transform--Heaviside Step Function, Ramp Function, Rectangle Function, Sigmoid Function, Sign, Square Wave, Triangle Function

RELATED WOLFRAM SITES

http://functions.wolfram.com/GeneralizedFunctions/UnitStep/,
http://functions.wolfram.com/GeneralizedFunctions/UnitStep2/

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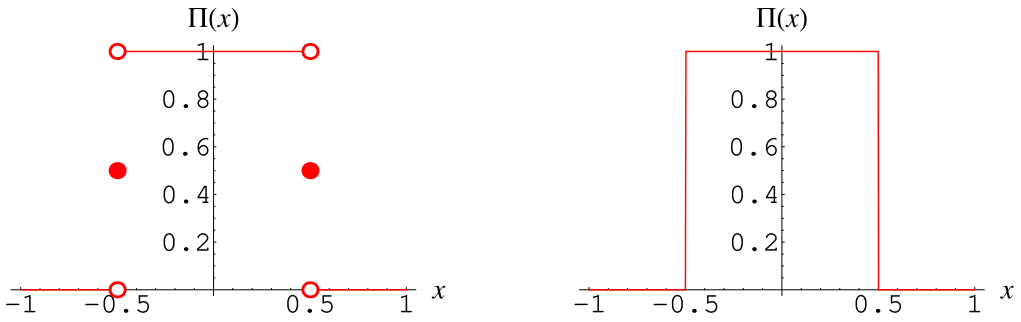
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Rectangle Function

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The rectangle function $\Pi(x)$ is a function that is 0 outside the interval $[-1/2,\ 1/2]$ and unity inside it. It is also called the gate function, pulse function, or window function, and is defined by

$$\Pi(x) \equiv \begin{cases} 0 & \text{for } |x| > \frac{1}{2} \\ \frac{1}{2} & \text{for } |x| = \frac{1}{2} \\ 1 & \text{for } |x| < \frac{1}{2}. \end{cases} \tag{1}$$

The left figure above plots the function as defined, while the right figure shows how it would appear if traced on an oscilloscope. The generalized function $f(x) = h\,\Pi((x-c)/b)$ has height h, center c, and full-width b.

As noted by Bracewell (1965, p. 53), "It is almost never important to specify the values at $x = \pm 1/2$, that is at the points of discontinuity. Likewise, it is not necessary or desirable to emphasize the values $\Pi(\pm 1/2) = 1/2$ in graphs; it is preferable to show graphs which are reminiscent of high-quality oscillograms (which, of course, would never show extra brightening halfway up the discontinuity)."

The piecewise version of the rectangle function is implemented in the Wolfram Language as UnitBox[x] (which takes the value 1 at $x = \pm 1/2$), while the generalized function version is implemented as HeavisidePi[x] (which remains unevalutaed at $x = \pm 1/2$).

Identities satisfied by the rectangle function include

$$\Pi(x) = H\left(x+\tfrac{1}{2}\right) - H\left(x-\tfrac{1}{2}\right) \tag{2}$$

$$= H\left(\tfrac{1}{2}+x\right) + H\left(\tfrac{1}{2}-x\right) - 1 \tag{3}$$

$$= H\left(\tfrac{1}{4}-x^2\right) \tag{4}$$

$$= \tfrac{1}{2}\left[\operatorname{sgn}\left(x+\tfrac{1}{2}\right) - \operatorname{sgn}\left(x-\tfrac{1}{2}\right)\right], \tag{5}$$

where $H(x)$ is the Heaviside step function. The Fourier transform of the rectangle function is given by

$$\mathcal{F}_x[\Pi(x)](k) = \int_{-\infty}^{\infty} e^{-2\pi i k x}\,\Pi(x)\,dx \tag{6}$$

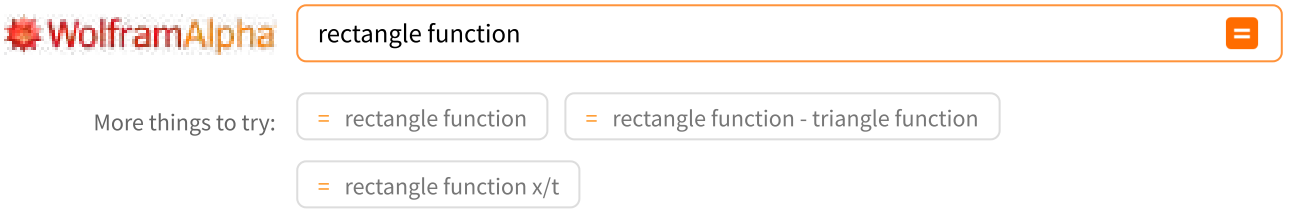
$$= \operatorname{sinc}(\pi k), \tag{7}$$

where $\operatorname{sinc}(x)$ is the sinc function.

SEE ALSO

Absolute Value, Boxcar Function, Fourier Transform--Rectangle Function, Heaviside Step Function, Ramp Function, Sign, Square Wave, Triangle Function, Uniform Distribution

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Bracewell, R. "Rectangle Function of Unit Height and Base, $\Pi(x)$." In *The Fourier Transform and Its Applications.* New York: McGraw-Hill, pp. 52-53, 1965.

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